Table 3: Search Model Counterfactual – What if No Search Costs

	Truth	NNE	SMLE ($\lambda = 3$)	SMLE ($\lambda = 7$)	SMLE ($\lambda = 10$)
Buy rate increase	0.141	0.135	0.161	0.173	0.168
	(.001)	(.001)	(.001)	(.001)	(.001)

Notes: Results are averaged across 100 Monte Carlo datasets. On each Monte Carlo dataset, we first estimate the search model and then use the estimated search model to simulate the counterfactual. Numbers in parentheses are standard errors for the averages.

We now turn to counterfactual analysis. We focus on a counterfactual that computes the impact of search cost on purchases. Specifically, given a value of $\boldsymbol{\theta}$, we can use the search model to compute the increment in buy rate if consumers no longer incur search costs (with everything else unchanged). This increment measures the purchases lost due to search costs. It provides the platform with an upper bound on the effect of interventions to reduce search costs (e.g., better web design, better ranking algorithms).

Table 3 displays the counterfactual results under the true $\boldsymbol{\theta}$, NNE, and SMLE. The increment in buy rate is closest to the truth when we use NNE's estimate. SMLE gives us a considerably higher increment than the truth. Intuitively, this is because search costs are over-estimated with SMLE (see Figure 5).

4.4 Real data estimation

We estimate the search model on a real dataset. Specifically, we use the Expedia hotel search data in the main analysis of Ursu (2018). There are $n = 1055$ search sessions (which can be treated as individual consumers in this search model). The number of options $J = 33$ for some consumers and $J = 34$ for the others. Table 4 reports the results.[16] We examine model fit later in Figure 8.

The top panel of Table 4 reports the computational costs, measured in terms of simulation burden and compute time. As defined before, simulation burden is the number of search model simulations required to carry out estimation, and compute time is total time used to carry out estimation by our machine and code. Overall, we see the computational cost of NNE is a number of times smaller than SMLE.

In the lower panel of Table 4, the first column reports the estimates by NNE. We see that the hotel star rating, location score, and promotion have positive effects on the consumer utility. The review score has a negative effect (as in Ursu 2018), though the estimate is not statistically significant. Price has a negative effect and it is statistically significant. As to search costs, the ranking effect (δ_1) is positive but not statistically significant. The lack of significance for δ_1 is likely because we allow a free search and there are only 8.8% (or 93) consumers who searched beyond the free search, which makes it statistically difficult to pin down how ranking affects search cost. Overall, the NNE's estimates are reasonable.

[16] Our model setup largely follows Ursu (2018) but there are a few differences, so the estimates are not directly comparable. Specifically, we assume free first search and take the logs of ranking and price. See Section 4.1.